Learning to Search the Literature: Budgeted Citation-Graph Agents for Systematic Review

Midway Report

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1 Research Questions

Systematic reviews depend on a search that finds every eligible study, and a missed study can change a review's conclusion. We study literature search as a budgeted sequential decision problem: given a review's question and eligibility criteria, an agent spends a fixed number of tool calls on keyword search, citation chasing and reading, and is scored on how many of the review's included studies it recovers. We ask four questions. (Q1) At equal budget, do prompted LLM agents beat retrieval and scripted snowballing baselines? (Q2) Can a small open model trained with supervised fine-tuning and reinforcement learning match or beat a much larger prompted model? (Q3) Which strategies emerge: how do agents divide their budget between search and citation chasing, and when do they stop? (Q4) Do the results transfer to reviews from other fields? Proposal feedback added a further question: is the citation graph itself limiting, and would open-web search help? The results raised one more: (Q5) can an agent tell how much it has missed, i.e. estimate its own recall when it stops?

2 Environment & Data

Data. Gold sets come from SYNERGY+ (De Bruin et al., 2023): 72 published systematic reviews, each with the candidate papers its authors screened and the subset they included (the gold set R). Metadata, abstracts and the citation graph come from OpenAlex (Priem et al., 2022) and are frozen into a local store, so episodes run offline and are reproducible. Splits are by review (Table 1). The 8 evaluation reviews are fixed for all systems and themselves split train/dev/test; the only tuned baseline parameter was chosen on the train reviews. The 24 transfer reviews come from fields that do not appear in training.

set	reviews	fields	used for
evaluation	8 (4 train / 2 dev / 2 test)	life science	all systems
training	40	life science	teacher trajectories, SFT, GRPO; extra evaluation of prompted systems
transfer	24	software eng., education, psychology, …	out-of-field evaluation

Table 1: Review splits. The sets are disjoint. The proposal also listed CLEF TAR and CSMeD; SYNERGY+ alone provided enough reviews, so we did not add them.

Environment. Each episode is a partially observed decision process. The agent observes the review question, the eligibility criteria, its remaining budget B, the latest tool result, a frontier of the 50 most recently seen papers, and its included and excluded sets. Episodes start from five seed papers found by BM25 (Robertson & Zaragoza, 2009) on the question and criteria. Each turn the agent may screen any paper it has seen (include or exclude, free) and take one paid action: search(query, k≤10), which costs 1 and returns the top unseen BM25 results (repeating a query pages deeper); expand_refs(p) and expand_citers(p), which cost

1 per 10 results ($k \leq 20$) and return a paper's references or citers; read(p), which costs 1 and returns the abstract; and stop. Budgets are $B \in \{10, 20, 40\}$, and there are three tool sets: keyword (search only), citation (expansion only) and hybrid (both). Once the budget is spent the agent may still screen what it has seen.

Leakage control. The review paper, every paper that cites it, and every paper published after it are hidden from the agent. Gold labels are used only by the scorer, never in prompts, seeds or queries, and an automated leakage audit was run on the training data before any training.

Search universes. To test whether the graph is limiting, the same agents run in three universes of increasing size: (i) the closed pool, the papers the review authors screened; (ii) the open graph, the pool plus up to 5,000 papers it cites, roughly doubling the distractors; and (iii) live web, all of OpenAlex (∼250M works), with search, references and citers served by the live API and cached for replay.

Metrics and success criteria. The primary metric is recall $|S \cap R|/|R|$ of the included set S at each budget. We also report precision and F1 of S, the area under the recall-versus-cost curve, and the action mix. Comparisons are paired by review, with 95% bootstrap confidence intervals over reviews and win/loss counts. A system "beats" another only when the interval excludes zero. The training reward is $r = |S \cap R|/|R| - \lambda\,|S \setminus R|/|S|$ with $\lambda = 0.1$.

3 Methods

Baselines. (1) BM25 and (2) SPECTER2 (Singh et al., 2023) rank the pool against the review text and include every result the budget pays for. This is the recall a retriever reaches without screening, at the lowest possible precision. (3) A scripted snowball agent searches, screens by keyword overlap, and chases the references and citers of included papers (Wohlin, 2014). (4) A random traversal.

Prompted agents. One prompt and one tool protocol for every model. Each turn the model returns JSON with screening decisions and exactly one action; invalid actions are charged and logged. Models: Claude Sonnet 5.5 and Haiku 5.5, and Qwen3-32B (Yang et al., 2025), at temperature 0 via OpenRouter, with a hard spending cap enforced in code.

Trained agent (focus of the investigation). Qwen3-1.7B with LoRA (Hu et al., 2022), trained in two stages after a pre-registered gate was met: at least 200 valid teacher trajectories (we had 360), a passing leakage audit, and a one-GPU smoke test. SFT: rejection sampling keeps the better half (by reward) of Haiku 5.5 trajectories on the 40 training reviews, giving 2,702 turns; 337 steps on one L40S. GRPO (Shao et al., 2024): starting from the SFT adapter, groups of 8 rollouts per task, KL penalty to SFT ($\beta = 0.02$), 40 steps at $B = 20$ over all three tool sets. Rollouts and evaluation are served with vLLM (Kwon et al., 2023). Both stages were repeated with a second seed. The open question for the rest of the quarter is what RL adds beyond imitation, and whether a trained policy learns to use citation tools at all.

4 Summary of Progress

Done.

- The environment, all baselines, the prompted agents and the training pipeline are implemented and tested (54 unit tests, including leakage and cost-accounting tests).
- Every system has been run on the 8 evaluation reviews × 3 tool sets × 3 budgets, in both the closed pool and the open graph, and on the 24 transfer reviews. The prompted systems were also run on the 40 training reviews, so their life-science comparisons rest on 48 reviews.
- Robustness checks:
  - three temperature-0.7 repeats for the API models;

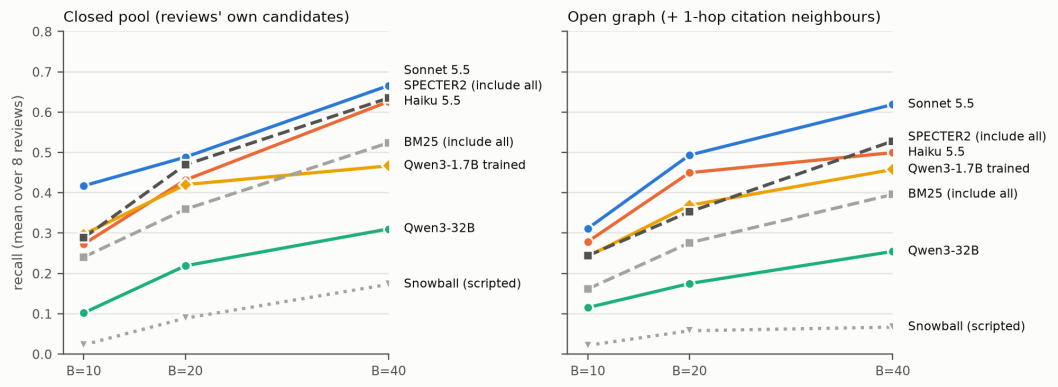


Figure 1: Recall by budget with hybrid tools, mean over the 8 evaluation reviews. Left: closed pool. Right: open graph. Dashed lines are retrieval baselines that include everything they see.

  - a closed-book memorization probe, with all systems rescored after removing every gold paper any model could name from memory;
  - a second training seed;
  - an agent that uses SPECTER2 as its search tool.
- A step-by-step HTML replay of episodes, for the demo.

Preliminary results. Recall at $B = 40$ with hybrid tools; brackets are 95% CIs of the paired difference.

- Q1. Sonnet beats BM25 by $+.152$ $[+.07, +.23]$ on 48 life-science reviews and by $+.087$ $[+.01, +.16]$ on the 24 transfer reviews. Its precision is about $8\times$ higher (.42 vs. .05).
  - The scripted snowball loses to Sonnet on 8 of 8 reviews at every budget.
  - Exhaustive SPECTER2 is the strongest baseline: its recall ties Sonnet's in the closed pool ($+.031$ $[-.05, +.12]$), but its precision is only .06.
- Q2. SFT moves the 1.7B model from .13 to .43 recall (closed pool); no review gets worse.
  - The trained model beats prompted Qwen3-32B by $+.133$ $[+.08, +.19]$ on transfer.
  - It matches its Haiku teacher out of field ($+.007$), but trails it in field ($-.159$).
  - GRPO adds $+.01$ to $+.07$ over SFT in all six seed $\times$ condition cells. Seed-to-seed variation is about as large.
  - GRPO v2 (F_2 reward, turn-level credit from each call's marginal gold gain with a redundancy penalty, in the spirit of PaSa's per-call reward (He et al., 2025), mixed budgets, 120 steps; two seeds plus an ablation without turn credit) missed its pre-stated criteria: at $B \geq 20$ no run reached precision $\geq .25$ with recall within .02 of v1, and its recall gain over SFT was not consistently beyond seed noise. Turn credit trades recall for precision (transfer, $B = 40$: recall $-.052$ $[-.10, -.01]$, precision $+.062$ $[+.05, +.08]$, F1 $+.042$ $[+.02, +.06]$), and with hybrid tools the policy still spends all its budget on search.
  - Exploratory: with citation tools only, v2 stops early less often (20–28 of 40 calls spent, vs. about 8 for SFT), and recall rises by $+.09$ to $+.24$ over SFT in all six seed $\times$ universe cells on 8 reviews. Absolute recall stays low (.13–.30).
- Q3. Given both tools, frontier agents spend 80–90% of the budget on new queries and almost none on citation chasing; the trained model re-pages a dozen queries. With citation tools only, every model stops early (Sonnet leaves half of $B = 40$ unused).

- Graph question. Opening the graph lowers every system's recall (Sonnet .67→.62, BM25 .52→.40), although 77–100% of gold lies within three hops of the seeds. The budget and the off-topic papers it must pay for limit the agents, not graph connectivity.
- Live web. On all of OpenAlex, Sonnet's recall falls from .67 to .32 and Haiku's from .63 to .17. Raw OpenAlex search finds 1% of gold, so the LLM's lead over include-all retrieval grows from +.14 (closed) to +.22 (open graph) to +.31 (web). On the 24 transfer reviews Haiku falls from .48 to .10 ($-.379$ $[-.47, -.30]$), still ahead of raw search (.02) by $+.080$ $[+.06, +.11]$.
- Forced citation chasing. Replacing every third action with expansion of a paper the agent already included lowers recall (Haiku $-.097$ $[-.19, -.02]$ closed, $-.071$ $[-.12, -.03]$ on 24 transfer reviews; Sonnet $-.018$, n.s.; trained 1.7B $-.064$ $[-.12, -.01]$ at $B = 20$, $-.003$ at $B = 40$). Unlike PaSa (He et al., 2025), in our setting citation hops are a poor use of budget.
- Q5. End-of-episode recall estimates rank episodes ($r \approx .5$–$.7$) but are inaccurate per episode (error $\approx .2$). An LLM screener cannot audit the agent: it accepts 92% of gold but also 87% of the agent's false inclusions.
- Checks. Removing memorized gold (7% of closed-pool gold) changes no system's recall by more than .02.

Still to do.

- Test whether larger budgets (100–400 calls) change the search-versus-citation result, and how close recall gets to the level real reviews need.
- Tighten the end-of-episode recall bound (mean lower bound is only about .15).
- Hand-label a sample of "false positives" to estimate how often agents find eligible studies missing from the gold set, as proposed.
- Write a fuller error analysis by failure type: premature stopping, reachable but not reached, screened out, missing abstract.
- Write the related-work section, the live demo, and a README quick-start that runs one episode on a small subgraph with the released checkpoint.
- Try the refutation-reward variant if time allows (stretch goal).

Obstacles and concerns.

- OpenAlex request limits. The key allows about 1,000 searches a day, so the live-web runs are paced across days.
- API spending. About $55 of the $60 project cap is spent, leaving little room for new frontier-model runs.
- Small in-field test set for the trained model. Its 40 training reviews cannot be reused, so its in-field claims rest on 8 reviews. Its transfer claims rest on 24.
- Training variance. Two seeds differ by up to .07 recall, as much as the GRPO effect, so ranking training recipes will need more seeds.
- Missing abstracts. About a quarter of gold papers lack an OpenAlex abstract, which caps screening accuracy for every system.

References

Jonathan De Bruin, Yongchao Ma, Gerbrich Ferdinands, Jelle Teijema, and Rens Van de Schoot. SYNERGY – open machine learning dataset on study selection in systematic reviews, 2023. URL https://doi.org/10.34894/HE6NAQ.

Yichen He, Guanhua Huang, Peiyuan Feng, Yuan Lin, Yuchen Zhang, Hang Li, and Weinan E. PaSa: An LLM agent for comprehensive academic paper search. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1:

Long Papers), pp. 11663–11679, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.572.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In International Conference on Learning Representations (ICLR), 2022.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In Proceedings of the 29th Symposium on Operating Systems Principles (SOSP), 2023.

Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. arXiv preprint arXiv:2205.01833, 2022.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval, 3(4):333–389, 2009.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. arXiv preprint arXiv:2402.03300, 2024.

Amanpreet Singh, Mike D'Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 5548–5566, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338.

Claes Wohlin. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE), 2014.

An Yang et al. Qwen3 technical report. arXiv preprint arXiv:2505.09388, 2025.

A AI-Use Disclosure Statement

[DRAFT: Katherine, check this against what you actually did and edit before submitting.] Ideation. The research questions and formulation come from my proposal, written by me with the limited assistance described in its own disclosure. Code and experiments. I used Claude Code (Anthropic), an AI coding agent, extensively. Under my direction it wrote most of the implementation: environment, baselines, LLM agent, training scripts, analysis scripts and tests. It also ran the experiments on Stanford FarmShare and computed the analyses and figures. I set the research questions, the evaluation protocol, the pre-registered training gate and the spending limits, and I reviewed the results. Writing. Claude Code drafted this report from the experiment logs. I revised it and am responsible for every claim in it. The LLM agents studied in this project (Claude Sonnet 5.5, Claude Haiku 5.5, Qwen3) are the objects of study, not authors.